



Alternating Optimization for LLM Post-Training: From Divergence to Verified Solutions

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ALS & its 3 repairs → **DIVERGE** | SGD / AdamW / LoRA → **CONVERGE** | proof below (matched budgets, pretrained baselines shown)

Research Question

Why does ALS-based post-training diverge on deep LLMs, and what works instead? ALS solves $W^* = (X^T X + \lambda I)^{-1} X^T Y$ in closed form — one solve replacing hundreds of gradient steps. It converges on ≤ 24 -layer models but diverges on every observed 28-layer attempt. Two sub-questions: (Q1) can the divergence be explained and fixed? (Q2) what actually works, compared fairly?

The 2×2 Framework — Four Post-Training Algorithms

Two design choices define how a model is fine-tuned: **which optimizer** drives the update, and **how many parameters** are trained. Crossing them gives four algorithms. Here is each one, in plain terms:

	Full-rank train ALL parameters	LoRA train a small low-rank adapter
AdamW standard gradient optimizer (Adam + weight decay)	B: gradient descent updates every weight of the whole model. The industry standard for fine-tuning.	D: same optimizer, but only a tiny adapter $\Delta W = BA$ ($r \ll d$) is trained; the original model is frozen. Far fewer parameters.
ASP alternating opt.: ALS closed-form solve + SGD + perturbation	A: each cycle, ALS replaces the output-layer weights with a closed-form least-squares solution, then SGD and noise follow. Diverges on deep models.	C: same alternating schedule applied to the low-rank adapter. Converges, but quality is an order of magnitude worse than AdamW.

All four run until they consume the **same total FLOPs** (one ALS solve = one AdamW step), with shared data, seeds, and evaluation — so the comparison is fair.

The Story in 4 Steps

STEP 1 — ALS diverges on deep models

The closed-form solve $W^* = (X^T X + \lambda I)^{-1} X^T Y$ converges on ≤ 24 -layer models but diverges on every observed 28-layer attempt (11/11 on Qwen2.5-7B; PPL → NaN in 1–3 cycles).

STEP 2 — Mechanism proven by controlled ablation

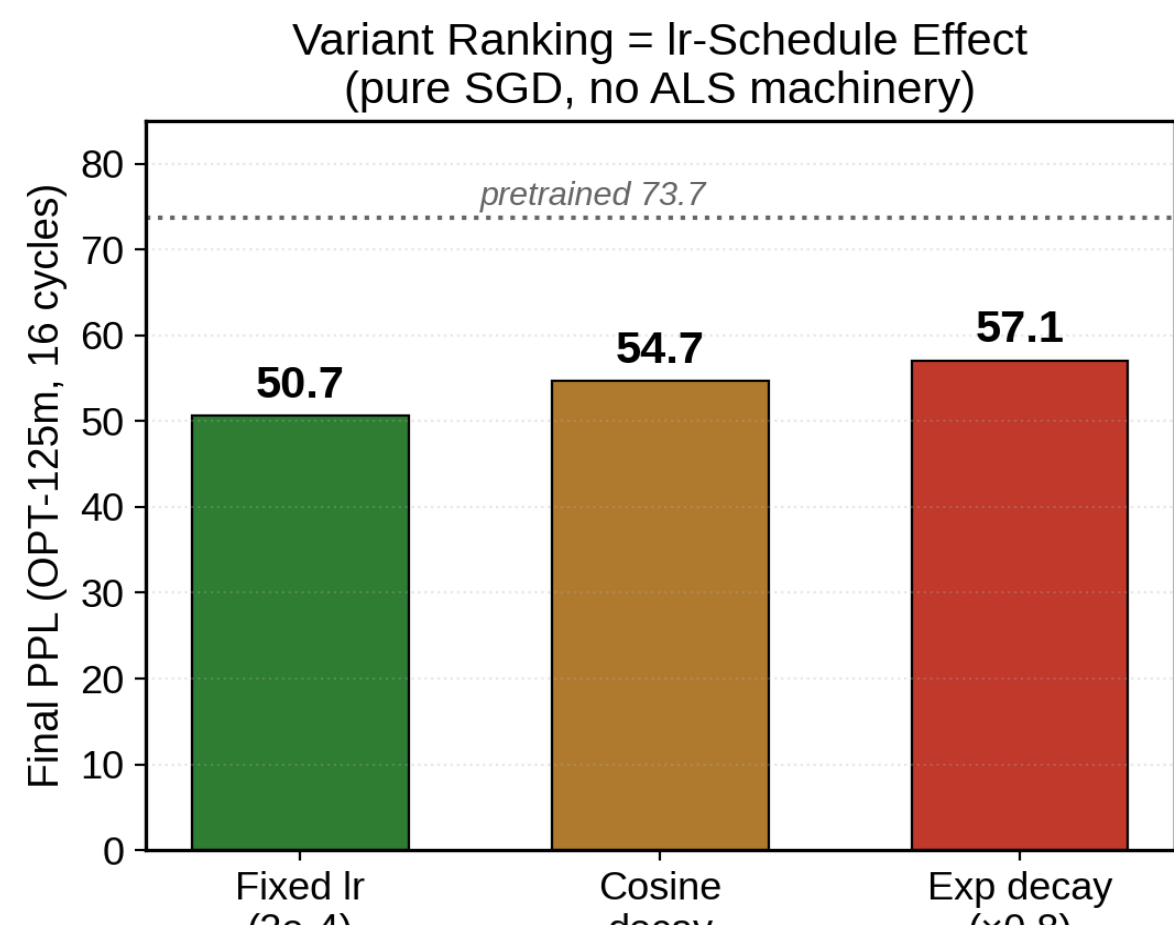
Five-condition ablation on Qwen2.5-7B isolates **ALS weight modification** as the necessary-and-sufficient cause within the tested space: every condition with a real weight solve diverges; SGD-only, perturbation-only and suppressed-weight ALS all converge (Fig. 5).

STEP 3 — All three natural repairs fail matched controls

Gradient injection, low-rank probing and soft-target distillation — each matched against an equal-budget SGD baseline — add no measurable value (Figs. 1–3).

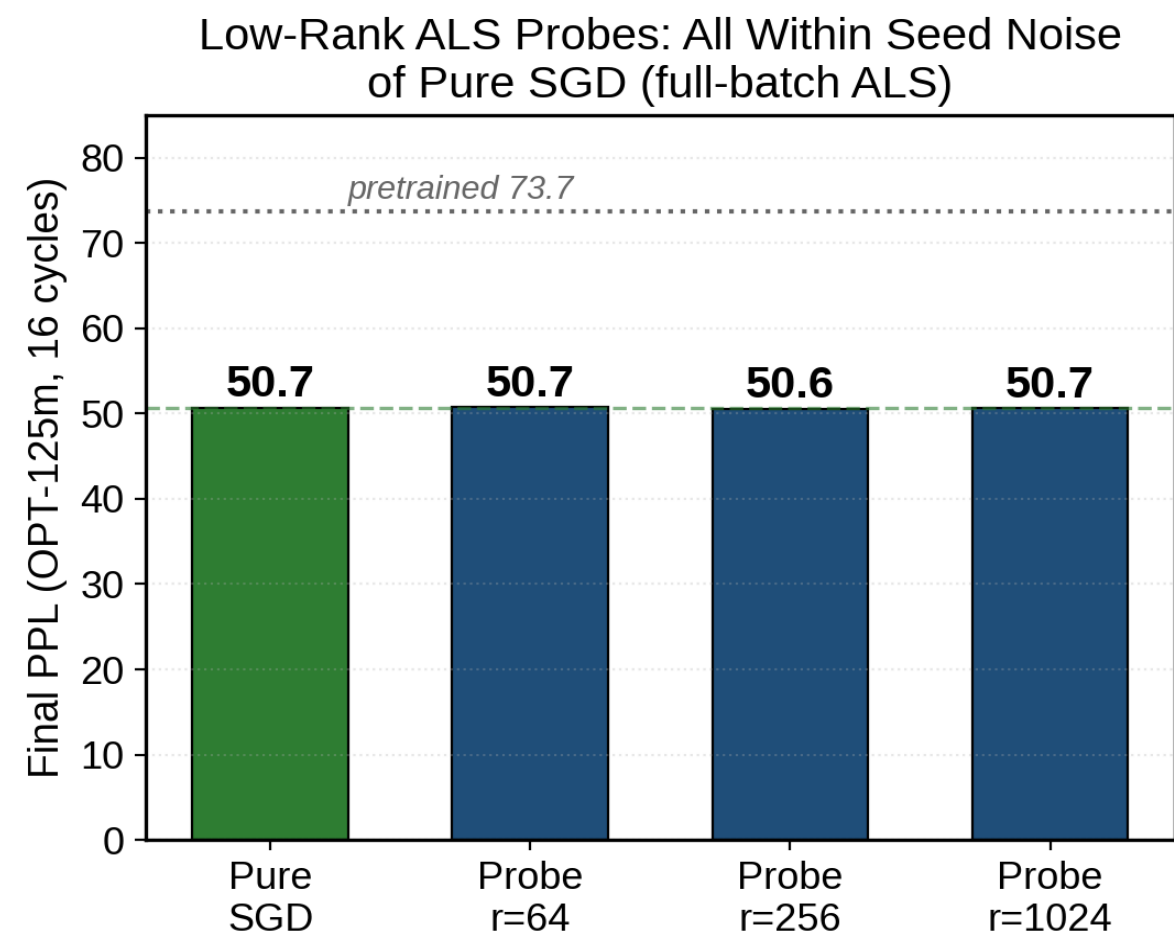
Repair analysis 1 — Variant ranking = learning-rate-schedule effect

With injection vacuous, the ranking is explained by the lr schedule alone (pure SGD, no ALS machinery): fixed 50.7 < cosine 54.7 < exp 57.1 — same order as the 7B variants.



Repair analysis 2 — Low-rank ALS probes: within seed noise at every rank

$r = 64 / 256 / 1024$ with full-batch ALS solves: 50.7 / 50.6 / 50.7 vs pure SGD 50.7 — no rank improves on SGD. The closed-form solve is redundant with the CE-gradient direction.



STEP 4 — Verified alternatives, baselines shown

AdamW full-rank 1.25 PPL; AdamW+LoRA 10.41 PPL at 0.1% trainable params; plain SGD 6.83 PPL. All far below the pretrained 73.1 PPL reference — **up to 58× improvement over doing nothing** (Fig. 4).

Contributions

- Diagnosis** — first controlled evidence that ALS weight modification (not noise / optimizer / hook overhead) causes deep-model divergence, via a 5-condition ablation.
- Theory** — residual-amplification model ($p \approx 1.08/\text{layer}$) predicts a 25–28-layer boundary, consistent with 8 architectures.
- Saved design space** — three plausible ALS repairs proven ineffective by matched-budget controls; identifies why (closed-form solve $\equiv$ CE gradient).
- Verified alternatives** — AdamW 58×, SGD 10.7×, LoRA 7× below the 73.1 pretrained baseline, with a corrected evaluation harness.

Why ALS diverges — causal theory

Residual connections amplify an ALS perturbation by an **effective $p \approx 1.08$ per layer** (phenomenological fit), i.e. $\approx 8\times$ over 27 layers — beyond SGD's per-cycle recovery. The predicted boundary falls **between 25 and 28 layers**, consistent with the observed 24-converges / 28-diverges split.

The perturbation *grows* across cycles (8: 0.085 → 0.196) — a positive feedback loop, which is why damping/clipping patches only delay the failure.

Method & discipline

- 2×2 factorial** (ASP vs AdamW × full-rank vs LoRA), FLOPs-matched, shared seeds/data.
- Harness audit:** OPT numbers re-verified with correct tokenizer + pad-masked labels (baseline 73.7 PPL); 7B baseline 73.1 PPL.
- Falsification discipline:** pre-stated hypothesis, open code, negative results reported with matched controls.

Verified Solutions (Qwen2.5-7B, 28 layers)

AdamW full-rank — 1.25 PPL (58× below baseline). Standard AdamW updates every weight of the model. It converges reliably at any depth and achieves the best absolute quality (N=3, 800 steps). *Use when:* quality is the priority and training compute is available.

AdamW + LoRA — 10.41 PPL (7× below baseline). Same optimizer, but only a tiny adapter $\Delta W = BA$ ($r=8$, $\sim 0.1\%$ of parameters) is trained; the base model is frozen. Nearly the same FLOPs as full-rank (backprop through frozen weights dominates), but the optimizer state and adapter memory are orders of magnitude smaller. *Use when:* storage/memory is constrained or many task-adapters must be served.

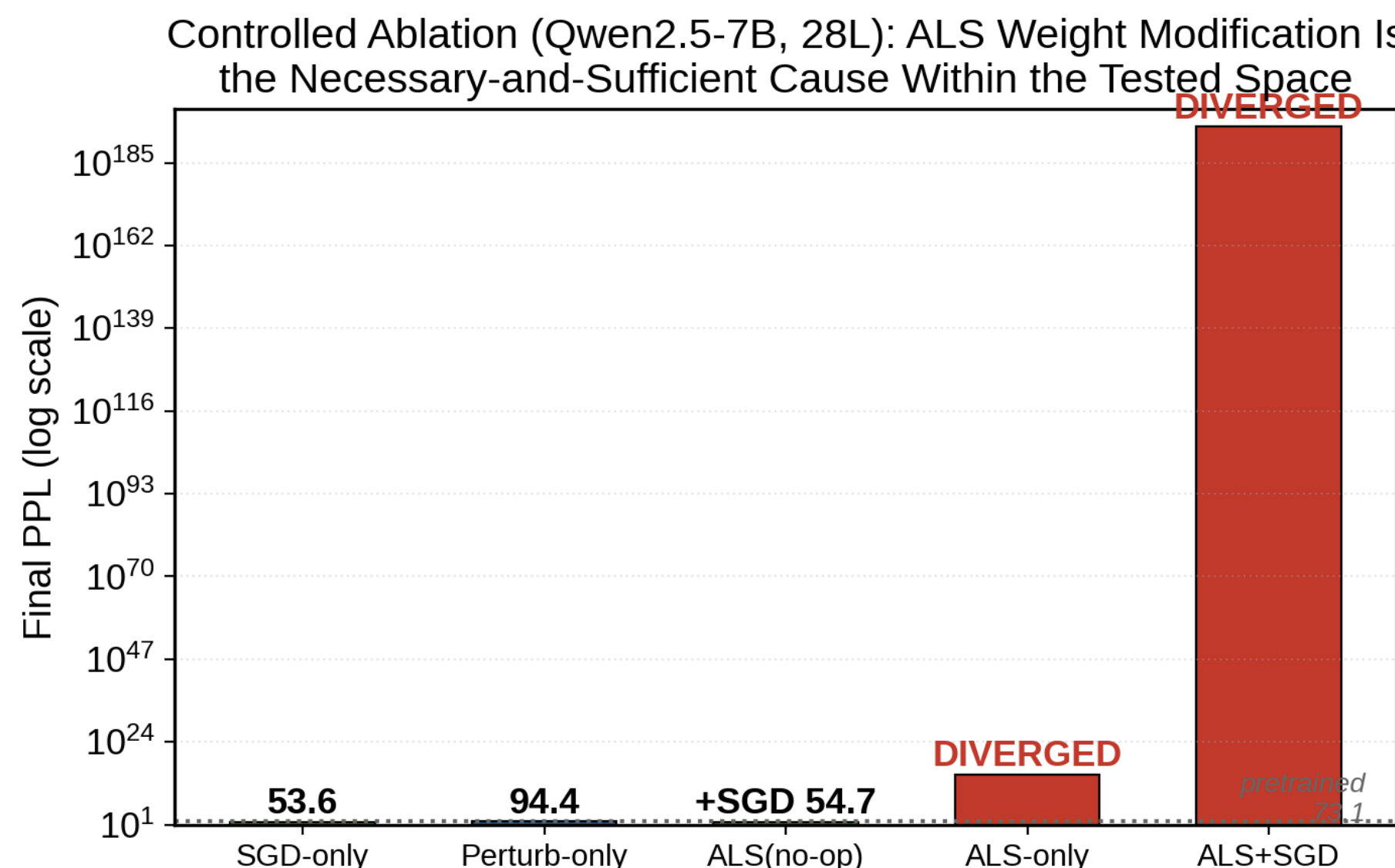
Plain SGD, fixed lr — 6.83 PPL (10.7× below baseline). Pure gradient descent with a sustained learning rate (no decay), no ALS machinery. It converges on 28 layers where ALS cannot — the decisive lesson that the failure lives in the ALS solve, not in gradient training. *Use when:* a simple, stable, dependency-free baseline is needed.

Method	Final PPL	vs 73.1 baseline	Best for
AdamW full-rank	1.25	58× better	absolute quality
Plain SGD (fixed lr)	6.83	10.7× better	simple stable baseline
AdamW + LoRA	10.41	7× better	memory / many adapters
ASP full-rank	DIVERGED	11/11 attempts	—

Key Findings & Experimental Results

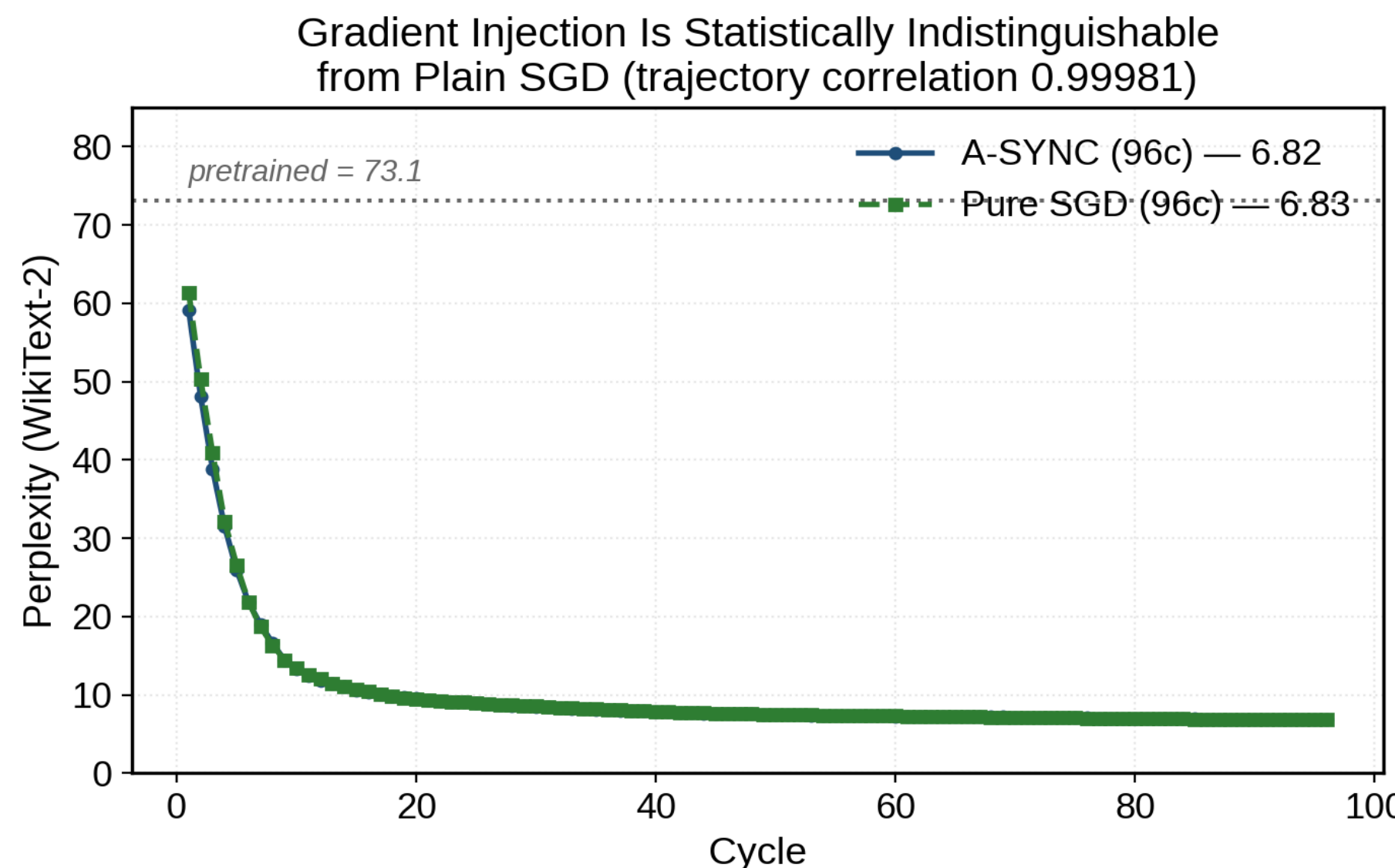
Finding 1 — ALS weight modification is the sole cause of divergence

A five-condition ablation on Qwen2.5-7B isolates the cause: every condition with a real weight solve diverges; SGD-only, perturbation-only and suppressed-weight ALS all converge on the same model.



Finding 2 — Gradient injection: statistically indistinguishable from plain SGD

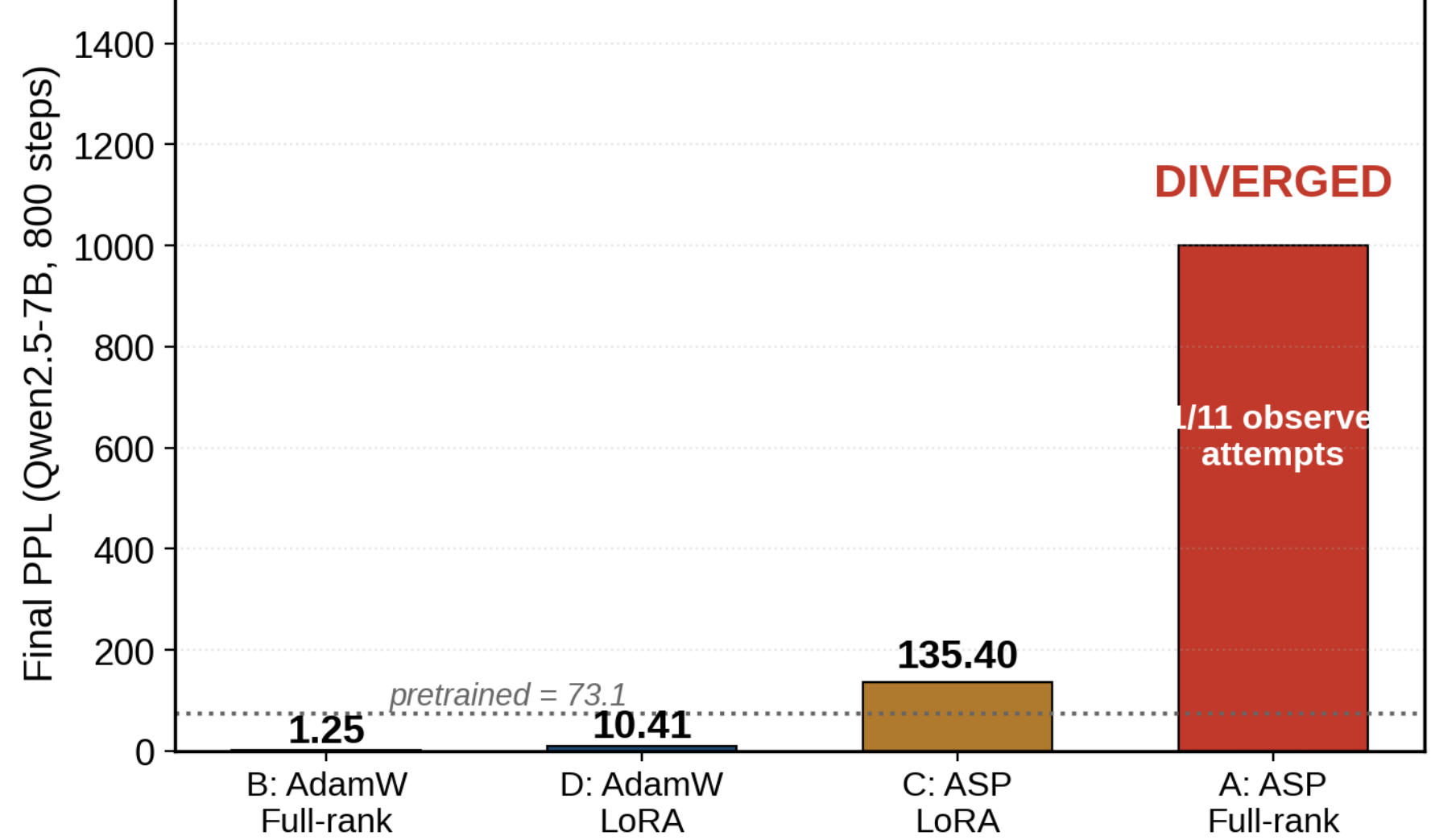
A-SYNC injects the ALS solution as a gradient bias; matched-budget pure SGD reproduces its trajectory (correlation 0.99981, mean per-cycle diff 0.116 PPL). No tested schedule beats plain SGD.



Finding 3 — The 2×2 matrix: AdamW below baseline, ASP diverges

On Qwen2.5-7B (800 steps, N=3) vs the 73.1 pretrained baseline: AdamW full-rank 1.25 PPL, AdamW+LoRA 10.41, ASP+LoRA 135.4, ASP full-rank diverges.

2×2 Factorial: AdamW Below Baseline; ASP Diverges Under Full-Rank



References & Reproducibility

- [1] Hu, E. et al. LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022.
- [2] Loshchilov, I., Hutter, F. Decoupled Weight Decay Regularization. ICLR 2019.
- [3] Koren, Y. et al. Matrix Factorization Techniques for Recommender Systems. IEEE Computer 42(8), 2009.
- [4] Merity, S. et al. Pointer Sentinel Mixture Models. ICLR 2017.
- [5] Hinton, G. et al. Distilling the Knowledge in a Neural Network. arXiv:1503.02531.

All experiments reproducible: runs/ + experiments/ in the project repo. Evaluation-harness audit and per-step data archived. Scan to view the repository:



Project repository
github.com/gingerseal/
alternating-optimization-lora

Takeaway: on deep models, remove the ALS machinery and let gradient descent do its job with a sustained learning rate — **AdamW 58× better than the pretrained baseline, plain SGD 10.7×, LoRA 7×**. Best quality still requires AdamW.

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